

TB-500 Available for Research Use Only

(2-2.5 mg 2×/wk for 4–6 wk, then 2-2.5 mg 1×/wk until healed)

TB-500 is widely used for tissue repair and injury recovery. Direct animal evidence makes dermal wound repair and tendon healing likely. Broader musculoskeletal, cardiovascular, and neurological repair effects are plausible based on animal evidence involving full-length thymosin β 4 (T β 4) and the retained LKKTETQ actin-binding region in TB-500. Documented veterinary use confirms animal exposure but does not independently establish efficacy.

 Possibly safe based on indirect thymosin β 4 evidence and very widespread community use of TB-500.

  Likely effective for dermal wound repair and tendon healing based on animal evidence. Broader tissue-repair benefits are plausible.

Cancer caution: TB-500's effects on angiogenesis and cell migration create a plausible concern that it could support the progression of an existing malignancy. Caution is appropriate for people with active or previous malignancy or otherwise elevated cancer risk.

Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.

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TB-500 is a synthetic N-acetylated heptapeptide, Ac-LKKTETQ, corresponding to residues 17 through 23 of the 43-amino-acid peptide T β 4. Public FRU testing reports commonly identify the Da Ac-LKKTETQ fragment as TB-500. Its retained sequence contains the principal actin-binding region associated with cytoskeletal regulation and cell migration.

How TB-500 Became a Top 10 Peptide

TB-500 grew out of legitimate research on T β 4, a naturally occurring 43-amino-acid peptide involved in actin regulation, cell migration, and tissue repair. **Researchers identified the short LKKTETQ (T β 4 17–23) region of T β 4** as biologically active and synthesized fragments containing this sequence for experimental study.

Clinical drug development largely pursued full-length T β 4, which entered human studies for several potential therapeutic applications. The much shorter fragment now commonly called TB-500 did not follow the same clinical-development path. **By the early 2010s, TB-500 had appeared in the veterinary and performance-enhancement markets and subsequently became widely available through research-peptide suppliers.**

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Its popularity grew rapidly, likely because it was **associated with the much larger Tβ4 literature** on wound healing, inflammation, angiogenesis, and tissue repair, and certainly because **TB-500 itself is comparatively simple to manufacture**. This combination helped turn TB-500 into one of the most widely used gray-market peptides, even though direct research on TB-500 itself remains much smaller than the research base for full-length Tβ4.

Dosing Patterns

No evidence-based human dose or dosing interval has been established for subcutaneous TB-500. Community protocols commonly use a **higher-dose initial loading phase** followed by **less intense maintenance dosing**. The following patterns are speculative community practices:

- Mild injury and general recovery: Loading, 2 mg twice weekly for 4 weeks. Maintenance, 2 mg once weekly during continued recovery.
- Musculoskeletal injury: Loading, 2.5 mg twice weekly for 4 to 6 weeks. Maintenance, 2 to 2.5 mg once weekly during continued recovery.
- Severe injury or cardiac recovery: Loading, 3 mg twice weekly for 6 weeks. Maintenance, 2.5 mg once weekly during continued recovery.

Sometimes twice-weekly dosing is divided into a similar weekly amount as daily doses.

Common Stacking

TB-500 is frequently combined with BPC-157 in community protocols. A 2026 controlled rat Achilles-tendon repair study found that TB-500 improved maximum load to failure and several histological measures, but combining it with BPC-157 did not provide additional benefit over either agent alone. Complementary biological effects remain plausible, while additive efficacy in humans is speculative.

Benefits

Most published research in this area studied full-length thymosin β4 rather than TB-500.

Evidence from full-length thymosin β4 studies supports plausible TB-500 effects when the proposed effect can reasonably be connected to the retained LKKTETQ actin-binding region. Direct animal evidence is narrower. Topically applied LKKTETQ promoted dermal wound repair in aged mice, and a 2026 rat Achilles-tendon repair study reported improved biomechanical and histological outcomes after injections of TB-500. These findings make dermal wound repair and tendon healing likely in the animal models studied. Detection studies also document administration of Ac-LKKTETQ to horses, but this veterinary use establishes exposure rather than therapeutic efficacy. Benefits for dermal wound repair and tendon healing from subcutaneous TB-500 in humans are plausible.

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- **Dermal wound repair:** TB-500 is likely to support dermal wound repair. Topical TB-500 promoted wound repair in aged mice, and effects on actin dynamics, cell migration, collagen deposition, and wound contraction provide plausible biological mechanisms.
- **Improved tendon healing:** TB-500 is likely to support tendon healing. In a 2026 rat Achilles-tendon repair study, injections of TB-500 improved maximum load to failure and several histological measures.
- **Cardiac tissue repair:** TB-500 plausibly supports cardiac cell survival, neovascularization, and post-infarction tissue remodeling based on animal evidence involving full-length thymosin β 4 and mechanistic effects associated with the retained LKKTETQ region.
- **Inflammatory signaling regulation:** TB-500 plausibly reduces neuroinflammatory signaling. In a 2026 study involving 5×FAD mice and cultured microglia, TB-500 administered with Ac-SDKP reduced glial activation, inflammatory microglial polarization, nitric oxide production, and pro-inflammatory cytokine expression. The individual contribution of TB-500 is uncertain. Broader anti-inflammatory effects are plausible.
- **Neurological tissue repair:** TB-500 plausibly supports neuronal survival, axonal integrity, neuroinflammatory regulation, and memory-related function. In a 2026 study involving 5×FAD mice and related cell models, TB-500 administered with Ac-SDKP reduced neurite atrophy, neuronal apoptosis, and glial activation, restored cortical axonal density, and improved memory-related behavioral performance. Benefits for traumatic brain injury and remyelinating conditions are plausible based on related animal evidence involving full-length thymosin β 4.
- **Hair-follicle support:** TB-500 plausibly supports hair-follicle stem-cell activity and hair growth based on animal evidence involving full-length thymosin β 4 and mechanistic activity associated with the retained LKKTETQ region.

Potential Applications

- **Acute soft-tissue repair:** TB-500 likely supports healing after tendon injury and dermal wounds based on animal evidence. Benefits following muscle or ligament injury and surgery are plausible.
- **Chronic overuse injuries:** TB-500 plausibly supports recovery from chronic tendinopathy and other overuse injuries through effects on cell migration, tissue remodeling, and inflammatory signaling.
- **Corneal surface repair:** A topical TB-500-containing hydrogel likely supports corneal epithelial regeneration in animal injury models. Corneal repair effects from subcutaneous TB-500 and benefits for dry eye disease or neurotrophic keratitis are plausible based on animal evidence for the fragment formulation and related human evidence involving full-length thymosin β 4.

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- **Post-infarction cardiac repair:** TB-500 plausibly supports cardiac cell survival, neovascularization, and tissue remodeling based on animal evidence involving full-length thymosin β 4 and mechanistic activity associated with the retained LKKTETQ region.
- **Neurological recovery:** TB-500 plausibly supports neuronal survival, axonal integrity, neuroinflammatory regulation, and memory-related function based on combined-peptide animal evidence. Benefits for traumatic brain injury and demyelinating conditions are plausible.
- **Hepatic fibrotic signaling:** TB-500 plausibly reduces hepatic stellate-cell activation, proliferation, migration, and collagen-related signaling based on in vitro research using LKKTETQ in human hepatic stellate cells.

Cautions

- **Active or previous malignancy:** TB-500's potential effects on angiogenesis and cell migration create a plausible concern that it could support the progression of an existing malignancy. **Active malignancy warrants particular caution.** For a previous malignancy, the relevance of this concern depends on current disease status, recurrence risk, and the biological features of the cancer.
- **Pregnancy and breastfeeding:** Reproductive, fetal, and lactation safety are unknown. Avoid use during pregnancy and breastfeeding as a precaution.
- **Children and adolescents:** Pediatric safety, dosing, and efficacy are unknown. Avoid use in children and adolescents outside qualified clinical or research oversight.

Side Effects

- **Community-reported effects:** Fatigue, headache, nausea, and transient flu-like symptoms are reported by some users. Their frequency and causal relationship to TB-500 are unknown.

Potential for Diminishing Benefit

TB-500 is not a classical receptor agonist, so receptor desensitization is not an expected primary mechanism of diminished response. As healing progresses, reduced inflammatory signaling, cell-migration demand, angiogenesis, and tissue-remodeling activity could plausibly reduce the marginal benefit of continued exposure. This would represent resolution-dependent diminishing benefit rather than demonstrated pharmacological tolerance. The value of cycling to prevent diminished response is speculative.

Biological Mechanism

Full-length thymosin β 4 binds globular actin through multiple regions of its sequence and regulates the balance between globular and filamentous actin. This supports cytoskeletal remodeling involved in cell shape, motility, migration, and tissue repair. TB-500 retains the

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central LKKTETQ actin-binding motif, making effects on actin-related cell migration and repair signaling plausible. The fragment should not be assumed to reproduce the full actin-sequestering activity of thymosin β 4.

Full-length thymosin β 4 contains several biologically active regions. Ac-SDKP corresponds to its N-terminal residues 1 through 4 and is associated with anti-inflammatory and antifibrotic activity. TB-500 instead reproduces the separate LKKTETQ actin-binding region at residues 17 through 23, which is associated with actin regulation, cell migration, angiogenesis, and wound repair. Evidence involving Ac-SDKP should not be attributed to TB-500.

The LKKTETQ actin-binding region retained in TB-500 promotes endothelial-cell migration and adhesion in vitro and vessel sprouting in an ex vivo chick-aortic model. These findings make angiogenic activity from TB-500 plausible. Increased local microvascular growth could plausibly improve oxygen and nutrient delivery during tissue repair.

TB-500 is a small peptide with a molecular weight of approximately 889 Da. Animal analytical studies demonstrate systemic exposure and formation of shorter metabolites after administration, and Ac-LKKTETQ has been detected in equine plasma and urine after injection. Distribution to distant injured tissues following subcutaneous administration is plausible.

TB-500 and Cancer Risk

TB-500 retains the LKKTETQ region associated with endothelial migration, vessel sprouting, and actin-related cell movement. Because existing tumors can exploit angiogenesis and cell-migration pathways, TB-500 could plausibly support the progression of an existing malignancy. Its effects on cancer initiation and long-term cancer risk are speculative. This mechanistic concern warrants particular caution in people with active malignancy and consideration in those with previous malignancy or elevated recurrence risk.

Conclusions

Safety is possible based on indirect thymosin β 4 evidence, animal tolerability, and very widespread community use of TB-500. TB-500's angiogenic and cell-migration activity creates a plausible concern that it could support the progression of an existing malignancy.

TB-500 is likely to support dermal wound repair and tendon healing based on animal evidence. Broader musculoskeletal, cardiovascular, neurological, and systemic tissue-repair benefits in humans are plausible.

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References

Direct TB-500 and LKKTETQ Evidence

1. Esposito S, et al. Synthesis and characterization of the N-terminal acetylated 17–23 fragment of thymosin beta 4 identified in TB-500, a product suspected to possess doping potential. *Drug Test Anal.* 2012;4(9):733–738. doi:10.1002/dta.1402.
2. Ho ENM, Kwok WH, Lau MY, et al. Doping control analysis of TB-500, a synthetic version of an active region of thymosin β 4, in equine urine and plasma by liquid chromatography-mass spectrometry. *J Chromatogr A.* 2012;1265:57–69. doi:10.1016/j.chroma.2012.09.043.
3. Rahaman KA, Muresan AR, Min H, et al. Simultaneous quantification of TB-500 and its metabolites in in-vitro experiments and rats by UHPLC-Q-Exactive orbitrap MS/MS and their screening by wound healing activities in-vitro. *J Chromatogr B Analyt Technol Biomed Life Sci.* 2024;1235:124033. doi:10.1016/j.jchromb.2024.124033.
4. Philp D, Badamchian M, Scheremeta B, Nguyen M, Goldstein AL, Kleinman HK. Thymosin beta 4 and a synthetic peptide containing its actin-binding domain promote dermal wound repair in db/db diabetic mice and in aged mice. *Wound Repair Regen.* 2003;11(1):19–24. doi:10.1046/j.1524-475x.2003.11105.x.
5. Philp D, Huff T, Gho YS, Hannappel E, Kleinman HK. The actin binding site on thymosin β 4 promotes angiogenesis. *FASEB J.* 2003;17(14):2103–2105. doi:10.1096/fj.03-0121fje.
6. Biçer O, Adanir O, Güleriyüz Y, et al. Effects of BPC-157 and TB-500 on Achilles tendon healing in rats: A histopathological and biomechanical study. *Jt Dis Relat Surg.* 2026;37(3):822–837. doi:10.52312/jdrs.2026.2951.
7. Ou Z, et al. Thymosin β 4-derived peptides alleviate neuroinflammation and neurite atrophy in in-vitro models and 5×FAD mice: A potential therapeutic approach for memory improvement in Alzheimer's disease. *Int Immunopharmacol.* 2026;116097. doi:10.1016/j.intimp.2025.116097.
8. Shah R, Reyes-Gordillo K, Rojkind M. Thymosin β 4 inhibits PDGF-BB-induced activation, proliferation, and migration of human hepatic stellate cells via its actin-binding domain. *Expert Opin Biol Ther.* 2018;18(Suppl 1):177–184. doi:10.1080/14712598.2018.1478961.
9. Lu P, Shan M, Peng C, Ji W, Yang T, Yang Z, Zhang Z, Wang Y. Alkaline phosphatase-triggered spatiotemporal repair of corneal injury with TB500 peptide hydrogel. *ACS Appl Mater Interfaces.* 2025;17(50):67503–67518. doi:10.1021/acsami.5c14652.

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Indirect Full-Length Thymosin β 4 Evidence

10. Bock-Marquette I, Saxena A, White MD, DiMaio JM, Srivastava D. Thymosin beta4 activates integrin-linked kinase and promotes cardiac cell migration, survival and cardiac repair. *Nature*. 2004;432(7016):466–472. doi:10.1038/nature03000.
11. Smart N, Risebro CA, Melville AAD, Moses K, Schwartz RJ, Chien KR, Riley PR. Thymosin beta4 induces adult epicardial progenitor mobilization and neovascularization. *Nature*. 2007;445(7124):177–182. doi:10.1038/nature05383.
12. Srivastava D, Ieda M, Fu J, Qian L. Cardiac repair with thymosin β 4 and cardiac reprogramming factors. *Ann N Y Acad Sci*. 2012;1270:66–72. doi:10.1111/j.1749-6632.2012.06696.x.
13. Xing Y, Ye Y, Zuo H, Li Y. Progress on the function and application of thymosin β 4. *Front Endocrinol (Lausanne)*. 2021;12:767785. doi:10.3389/fendo.2021.767785.

Thymosin β 4 — wound and tissue repair

14. Malinda KM, Sidhu GS, Mani H, Banaudha K, Maheshwari RK, Goldstein AL, Kleinman HK. Thymosin beta4 accelerates wound healing. *J Invest Dermatol*. 1999;113(3):364–368. doi:10.1046/j.1523-1747.1999.00708.x.
15. Philp D, Nguyen M, Scheremeta B, St-Surin S, Villa AM, Orgel A, Kleinman HK, Elkin M. Thymosin beta4 increases hair growth by activation of hair follicle stem cells. *FASEB J*. 2004;18(2):385–387. doi:10.1096/fj.03-0244fje.
16. Sosne G, Szliter EA, Barrett R, Kernacki KA, Kleinman HK, Hazlett LD. Thymosin beta 4 promotes corneal wound healing and decreases inflammation in vivo following alkali injury. *Exp Eye Res*. 2002;74(2):293–299. doi:10.1006/exer.2001.1125.
17. Sosne G, Ousler GW. Thymosin beta 4 ophthalmic solution for dry eye: a randomized, placebo-controlled, Phase II clinical trial conducted using the controlled adverse environment (CAE™) model. *Clin Ophthalmol*. 2015;9:877–884. doi:10.2147/OPTH.S80954.

Thymosin β 4 — neurological repair

19. Xiong Y, Zhang Y, Mahmood A, et al. Neuroprotective and neurorestorative effects of thymosin beta4 treatment initiated 6 hours post injury following traumatic brain injury in rats. *J Neurosurg*. 2012;116(5):1081–1092. doi:10.3171/2012.1.JNS111729.
20. Morris DC, Chopp M, Zhang L, Lu M, Zhang ZG. Thymosin β 4 improves functional neurological outcome in a rat model of embolic stroke. *Neuroscience*. 2010;169(2):674–682. doi:10.1016/j.neuroscience.2010.05.017.

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Authoritative Regulatory Assessment

21. U.S. Food and Drug Administration. Evaluation of TB-500-Related Bulk Drug Substances for Inclusion on the 503A Bulk Drug Substances List. FDA Evaluation of TB-500-Related Bulk Drug Substances (TB-500 Free Base and TB-500 Acetate). May 15, 2026. Pharmacy Compounding Advisory Committee briefing document for the July 23–24, 2026 meeting. <https://www.fda.gov/media/193349/download>.

Gray-Market Analytical Example

22. Janoshik. Test Report, Task No. 91865. Sample marketed as “TB-500 Frag 12-23 10 mg”; analytical result reported as “TB-500 (889 Da).” Analysis completed December 8, 2025. <https://www.obsidianpeptides.com/coas/tb-500.pdf>.